

Value Range Queries 1

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 256 megabytes

PPU Code Academy is preparing a training camp for its students. Before dividing students into groups, the organizers recorded the score of each student in a qualification test.

For different activities, the organizers want to know **how many students have scores within a specific range**.

For each query, you are given two values L and R . Count how many students have a score between L and R , inclusive.

Input

The first line contains two integers n and q ($1 \leq n, q \leq 10$), the number of students and the number of queries. The second line contains n integers a_i ($1 \leq a_i \leq 10$), where a_i is the score of the i -th student. Each of the next q lines contains two integers L and R ($1 \leq L \leq R \leq 10$), representing a score range.

Output

For each query, print the number of students whose score is between L and R , inclusive, seperated by a space.

Example

standard input	standard output
6 5 1 2 3 3 4 5 2 6 1 4 1 1 3 3 3 5	5 5 1 2 4